GOSIM

GOSIM Paris 2026

From LLM Apps to Agent Systems: Building an AI Operating System with Octos

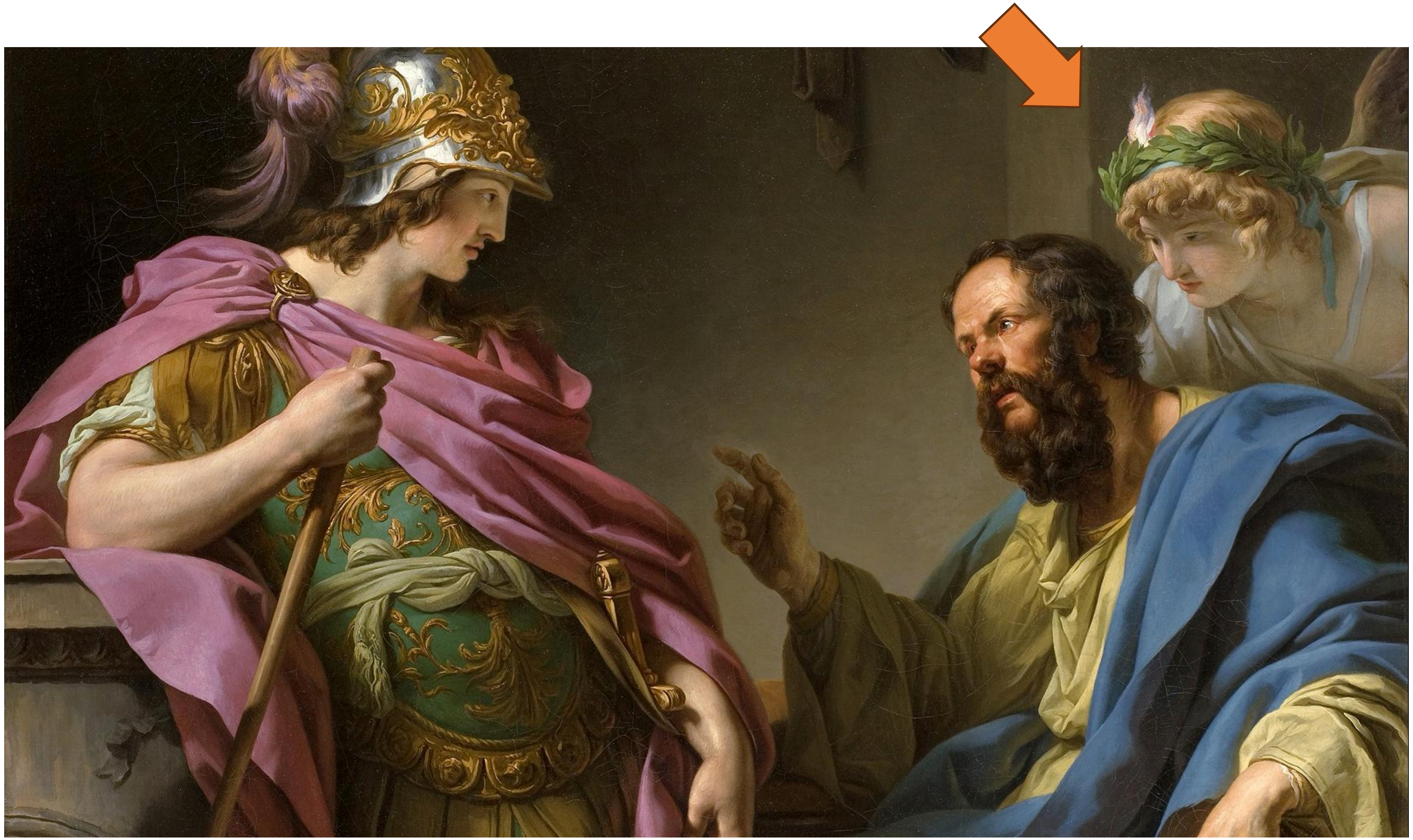
Zonghuan Wu

Octos

Toward Agent OS

Daemon

δαίμων
Daimon

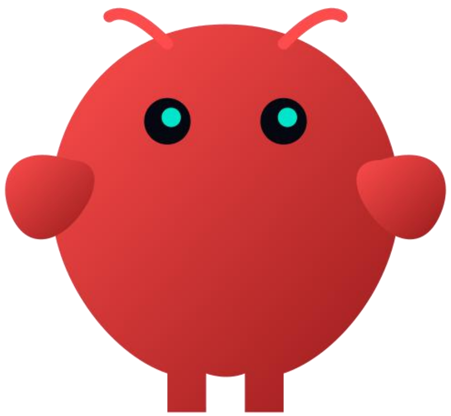


WIKIPEDIA
Den fria encyklopedin

Socrates with Alcibiades and the Daimonion. Oil painting by François-André Vincent, 1776, in the Musée Fabre, Montpellier

Assistant → Daimon

- More intelligent
- More Capable
- More Autonomous
- More Lasting

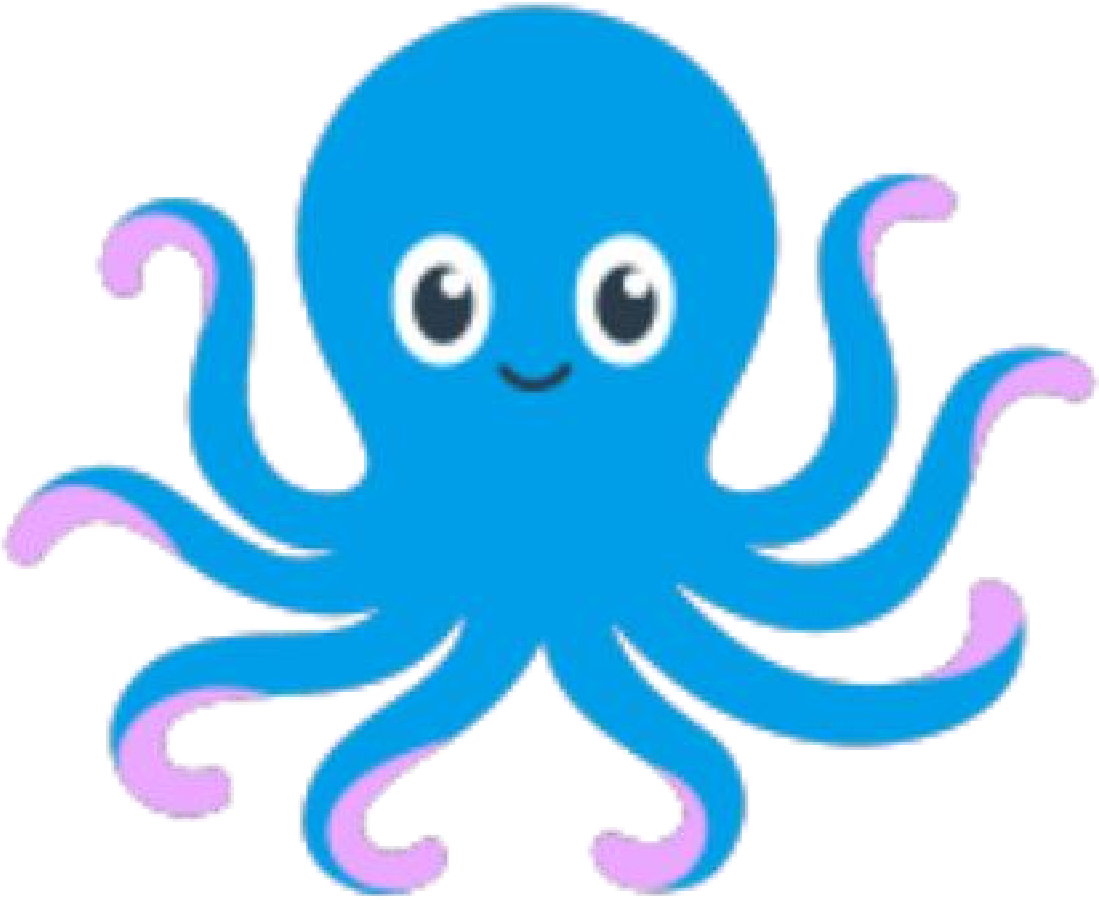


One Diamon	A fleet of daimons
OpenClaw, Hermes, and many claws …	octos
Personal	Personal, Team & Family
One user, one machine	One machine, one fleet
Skills bolted on per install	Skills curated, sandboxed, isolated

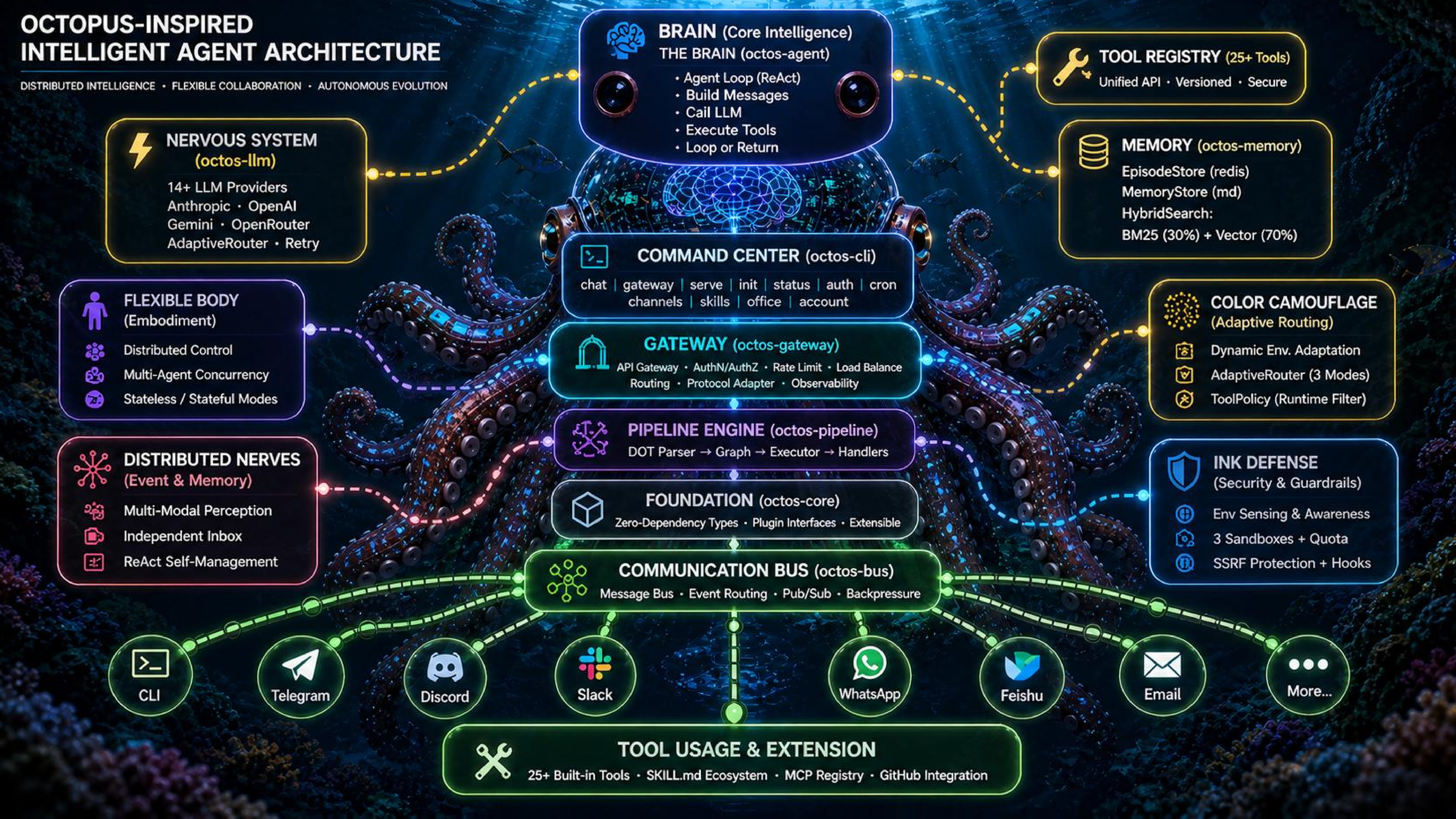
Octos

An OS for daimons

https://github.com/octos-org/octos



31 MB | pure Rust | single static binary | no Python, node, or Open SSL



OCTOPUS-INSPIRED INTELLIGENT AGENT ARCHITECTURE
DISTRIBUTED INTELLIGENCE • FLEXIBLE COLLABORATION • AUTONOMOUS EVOLUTION
BRAIN (Core Intelligence)
THE BRAIN (octos-agent)
• Agent Loop (ReAct)
• Build Messages
• Cail LLM
• Execute Tools
• Loop or Return
TOOL REGISTRY (25+ Tools)
Unified API • Versioned • Secure
NERVOUS SYSTEM (octos-llm)
14+ LLM Providers
Anthropic • OpenAI
Gemini • OpenRouter
AdaptiveRouter • Retry
MEMORY (octos-memory)
EpisodeStore (redis)
MemoryStore (md)
HybridSearch:
BM25 (30%) + Vector (70%)
COMMAND CENTER (octos-cli)
chat | gateway | serve | init | status | auth | cron
channels | skills | office | account
FLEXIBLE BODY (Embodiment)
Distributed Control
Multi-Agent Concurrency
Stateless / Stateful Modes
GATEWAY (octos-gateway)
API Gateway • AuthN/AuthZ • Rate Limit • Load Balance
Routing • Protocol Adapter • Observability
COLOR CAMOUFLAGE (Adaptive Routing)
Dynamic Env. Adaptation
AdaptiveRouter (3 Modes)
ToolPolicy (Runtime Filter)
PIPELINE ENGINE (octos-pipeline)
DOT Parser → Graph → Executor → Handlers
DISTRIBUTED NERVES (Event & Memory)
Multi-Modal Perception
Independent Inbox
ReAct Self-Management
FOUNDATION (octos-core)
Zero-Dependency Types • Plugin Interfaces • Extensible
INK DEFENSE (Security & Guardrails)
Env Sensing & Awareness
3 Sandboxes + Quota
SSRF Protection + Hooks
COMMUNICATION BUS (octos-bus)
Message Bus • Event Routing • Pub/Sub • Backpressure
CLI
Telegram
Discord
Slack
WhatsApp
Feishu
Email
More...
TOOL USAGE & EXTENSION
25+ Built-in Tools • SKILL.md Ecosystem • MCP Registry • GitHub Integration

1 · Multi-tenancy: a four-layer hierarchy

```
Profile / Account   →  LLM contract · tools · channels · env · skills · data dir
        ↓
Process             →  optional gateway subprocess per profile (process–manager mode)
        ↓
User                →  per–user workspace + sessions  (channel:chat_id is the user)
        ↓
Session             →  per–session ToolRegistry · JSONL history · cancel flag
```

Not a tenant-ID field. A composed isolation model.

User layer: dual enforcement

Application level — `resolve_path()` confines every tool's file ops to `workspace/`
Kernel level — macOS `sandbox-exec` SBPL profile pins file access to the same path

```
{data_dir}/users/{channel:chat_id}/
├── workspace/           # tool cwd, sandbox cwd
└── sessions/
    ├── default.jsonl
    └── research.jsonl
```

Each session actor gets its own sandbox.

Even with a buggy tool in the same Gateway process, user A's shell can't read user B's files — the OS rejects it.

What we don't claim

- ✅ Profile / Account: LLM, tools, channels, env, skills, data dir — isolated
- ✅ User: workspace + sessions — isolated, dual-enforced
- ✅ Session: ToolRegistry, history, cancel — isolated
- ⚠️ Process: only in process-manager mode; standalone Gateway is in-process actors
- ⚠️ Long-term memory: still at profile/global `data_dir/memory/`, not per-user
- ❌ Resource (CPU/RAM): **not container-level**. One profile under load can affect siblings.

> *If you need hostile multi-tenancy, run one octos per container. That's a deployment choice — not what we ship.*

Sub-accounts: shared contract, isolated edges

Parent — shared capability contract

LLM · search · deep_crawl · apps · email · env_vars (base)

Sub-account — interface + selective override

- own channels · own `public_subdomain`
- own `env_vars` (override parent)
- own `data_dir/skills/` ← **never inherited**

```
parent--child
sales--europe · sales--apac · support--enterprise
```

Why customer-installed skills don't inherit

```
parent installs:        ./skills/some-tool

sub-account inherits:   LLM, search, email, apps, env_vars  ✓
                        skills                              ✗
```

Security boundary. Sub-accounts run on different channels with different trust levels. Silent skill inheritance would expand blast radius across every customer-facing surface.

> *Not a config switch. An architectural floor.*
> *(`skills_scope.rs` resolves account skills strictly to the current account.)*

2 · The Adaptive Router

Three modes — picked **per route**:

```
off      sequential failover          predictable, table stakes
lane     composite score              latency · errors · priority · cost
hedge    race two providers           first wins, loser dropped mid-flight
```

Live metrics, shared across all modes:

EMA latency · p95 (64-sample circular buffer) · error rate · circuit state

```
                      ┌──→  Provider A  ──┐
    query  ──→   hedge                    ├──→  first wins
                      └──→  Provider B  ──┘   (loser dropped mid-flight)
```

3 · Pipelines: Agents, not function calls

Most pipeline engines — LangGraph · CrewAI · AutoGen · n8n
node = function call · agent loop, tool calling, model routing re-implemented per node

octos pipelines
node = full `octos_agent::Agent` · inherits tools, loop, model router, sandbox

Per-node `tools=` is an allowlist on top of a hard safety floor:
`spawn` · `run_pipeline` · `send_file` · `message` — always denied (prevents recursive runaway).

> **Composition, not reinvention.**

DynamicParallel — Agents that plan more Agents

```
plan [handler="dynamic_parallel",
      planner_model="strong",
      worker_prompt="research angle: {task}",
      model="cheap,strong,cheap",        ← worker pool, round-robin
      max_tasks="6",
      converge="synthesize"]
```

1. **Planner Agent runs first** — returns a JSON task list
2. **Executor synthesizes one worker Agent per task** — at runtime, from JSON
3. **Workers rotate through the model pool** — different cost/quality per worker
4. **Results merge at `converge`**

Runtime routing — branch on outcomes, not on code

```
test   -> deploy   [condition="outcome.status == \"pass\""]
test   -> rollback [condition="outcome.status == \"fail\""]
report -> refine   [condition="outcome.contains(\"missing data\")"]
```

Edge selection priority:

conditions → node's `suggested_next` → label match in outcome → weighted fallback

15 validation rules at parse time catch structural problems before runtime — no start, cycles, missing `converge`, unparseable conditions.

> **Validated upfront. Routed at runtime.**

What it enables: deep research

Task: *"Impact of the EU AI Act on European startups — write a 5-page report"*

	LangGraph · CrewAI · n8n	octos pipelines
Worker count	hard-coded up front	**planner decides at runtime (3–8)**
LLM call reliability	sequential failover → slow tail	**hedge mode races two providers**
Model selection	same model for all workers	**pool rotates: cheap · strong · cheap · strong**
Branch on outcomes	Python conditionals around graph	`condition="outcome.contains(...)"` **in DOT**

> **All four enabled by composition: router + pipeline + tool registry, working together.**

OCTOS WORKSPACE

Chat History

+ New chat

RECENT SESSIONS

Deep research the top...
LIVE SESSION

Deep research the top o...
LIVE SESSION

Deep research the top o...
SAVED SESSION

deepseek@api/deepseek-chat

admin@localhost

CURRENT SESSION

Deep research the top open-source AI agent framewo

Deep research running
Background work continues independently of chat messages.

deepseek-chat

Deep research the top open-source AI agent frameworks in 2026. Compare architecture, community adoption, and unique capabilities. Write a structured report.

2026-05-02 11:57:33

Deep research started in the background. The final report will be delivered to this session when complete.

```
Deep research
  Deep research running
```

2026-05-02 11:57:33

Send a message...

OCTOS WORKSPACE

Chat History

+ New chat

RECENT SESSIONS

Deep research the top...
SAVED SESSION

Deep research the top o...
SAVED SESSION

Deep research the top o...
SAVED SESSION

Deep research the top o...
SAVED SESSION

deepseek@api/deepseek-chat

admin@localhost

CURRENT SESSION

Deep research the top open-source AI agent framewo

deepseek-chat

Deep research the top open-source AI agent frameworks in 2026. Compare architecture, community adoption, and unique capabilities. Write a structured report.

2026-05-02 11:59:20

Deep research started in the background. The final report will be delivered to this session when complete.

```
Deep research
  Deep research running
  Deep research completed
```

2026-05-02 11:59:20

Status: SUCCESS

The research is complete. The full structured report has been saved to `` `ai_agent_frameworks_2026_report.md` ``.

Here is the executive summary:

Executive Summary: Open-Source AI Agent Frameworks in 2026

Key Findings

The open-source AI agent framework landscape in 2026 is defined by six major players — LangGraph, CrewAI, AutoGen, Semantic Kernel (Microsoft Agent Framework), Dify, and Haystack — each occupying a distinct niche. No single framework dominates all use cases; the optimal choice depends on specific requirements

Send a message...

SESSION FILES

Deep research the top ...

YWIfYWdlbnRfZnJhbWV3b3Jrc18yMD... 1

ai_agent_frameworks_202...
29.5 KB 5月2日 12:26 REPORT

OCTOS WORKSPACE

Chat History

+ New chat

RECENT SESSIONS

Deep research the top...
SAVED SESSION

Deep research the top o...
SAVED SESSION

Deep research the top o...
SAVED SESSION

Deep research the top o...
SAVED SESSION

deepseek@api/deepseek-chat

admin@localhost

CURRENT SESSION

Deep research the top open-source AI agent framewo

deepseek-chat

- Business process automation: CrewAI (enterprise integrations, role-based design)
- Rapid chatbot prototyping: AutoGen (low-code web interface)
- Enterprise Copilot development: Semantic Kernel (Microsoft ecosystem)
- RAG-powered knowledge bases: Dify (integrated RAG, visual workflow)
- Production search systems: Haystack (proven at scale)
- Multi-agent research systems: AutoGen or CrewAI
- Small startups (rapid MVP): Dify or AutoGen
- Enterprise IT departments: Semantic Kernel
- AI research labs: LangGraph

Conclusions

The open-source AI agent framework ecosystem in 2026 is mature and diverse, with clear leaders in each category. Organizations should evaluate frameworks based on their specific requirements rather than seeking a one-size-fits-all solution. For maximum flexibility, consider hybrid approaches — combining LangGraph for core orchestration with Dify for RAG pipelines, or Haystack for document processing with CrewAI for multi-agent coordination. The rapid pace of innovation across all six frameworks suggests continued evolution, with convergence toward better state management, enterprise integration, and production tooling across the board.

The full report with detailed framework deep dives, comparison tables, architecture diagrams, and sourced references is available in `ai_agent_frameworks_2026_report.md`.

ai_agent_frameworks_2026_report.md

2026-05-02 12:26:36

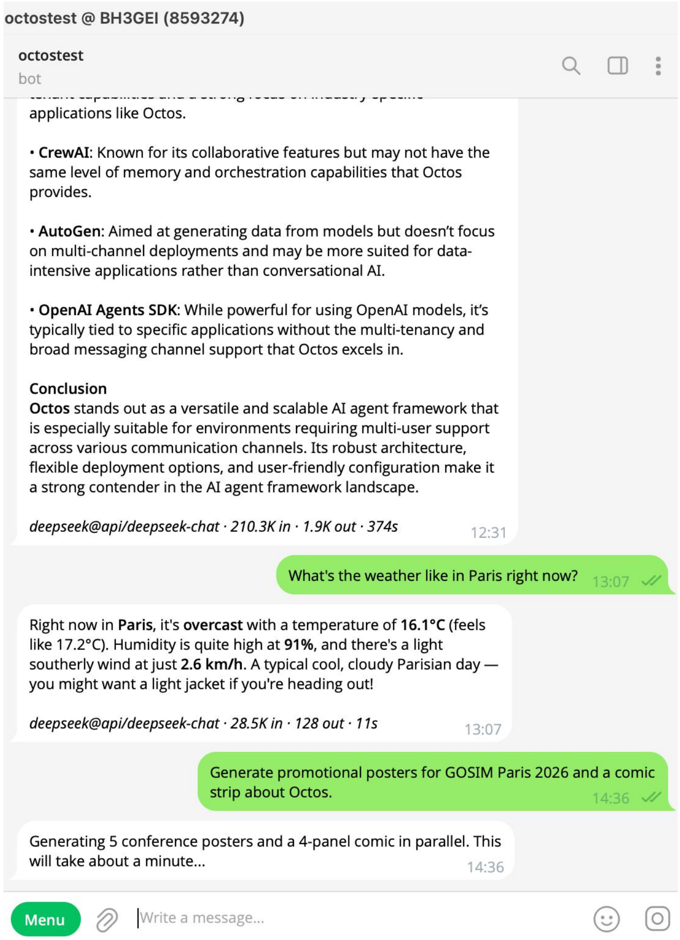
Send a message...

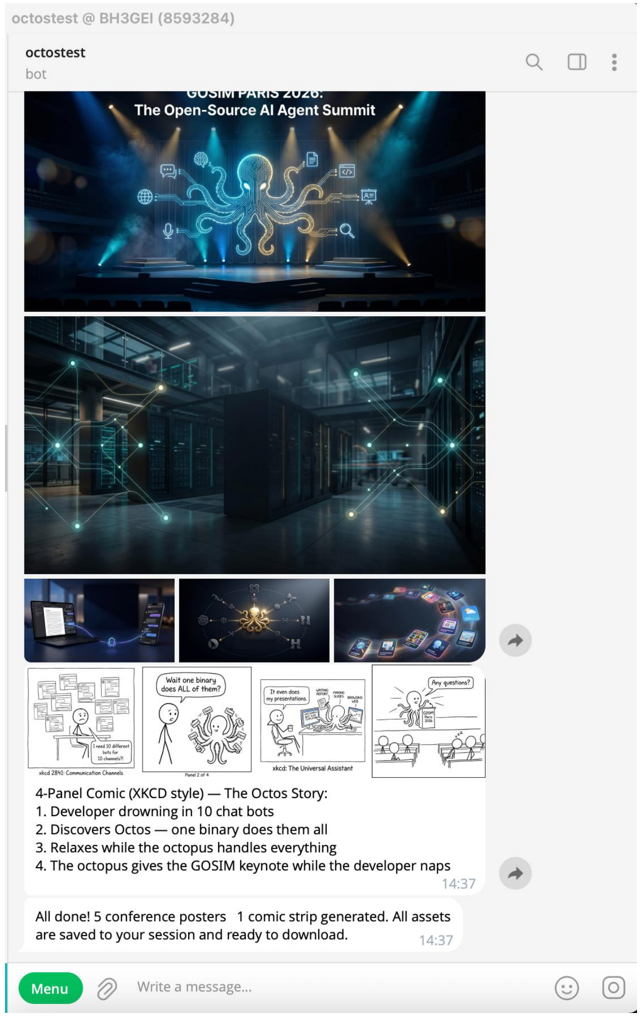
SESSION FILES

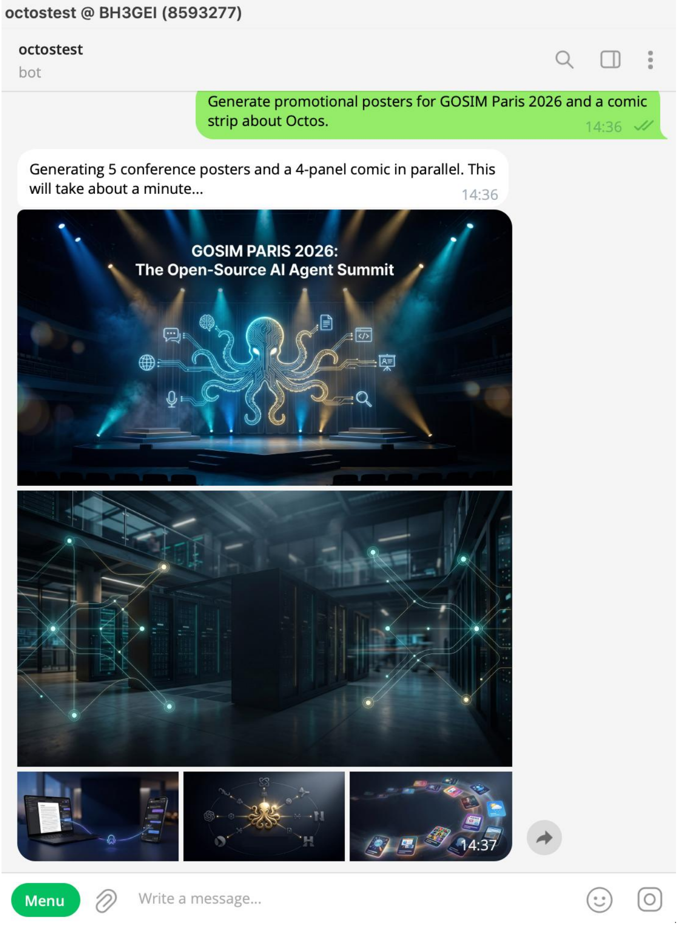
Deep research the top ...

YWlfYWdlbnRfZnJhbWV3b3Jrc18yMD... 1

ai_agent_frameworks_202...
29.5 KB 5月2日 12:26 REPORT


octostest @ BH3GEI (8593274)
octostest
bot
applications like Octos.
• CrewAI: Known for its collaborative features but may not have the same level of memory and orchestration capabilities that Octos provides.
• AutoGen: Aimed at generating data from models but doesn't focus on multi-channel deployments and may be more suited for data-intensive applications rather than conversational AI.
• OpenAI Agents SDK: While powerful for using OpenAI models, it's typically tied to specific applications without the multi-tenancy and broad messaging channel support that Octos excels in.
Conclusion
Octos stands out as a versatile and scalable AI agent framework that is especially suitable for environments requiring multi-user support across various communication channels. Its robust architecture, flexible deployment options, and user-friendly configuration make it a strong contender in the AI agent framework landscape.
deepseek@api/deepseek-chat · 210.3K in · 1.9K out · 374s
12:31
What's the weather like in Paris right now?
13:07
Right now in Paris, it's overcast with a temperature of 16.1°C (feels like 17.2°C). Humidity is quite high at 91%, and there's a light southerly wind at just 2.6 km/h. A typical cool, cloudy Parisian day — you might want a light jacket if you're heading out!
deepseek@api/deepseek-chat · 28.5K in · 128 out · 11s
13:07
Generate promotional posters for GOSIM Paris 2026 and a comic strip about Octos.
14:36
Generating 5 conference posters and a 4-panel comic in parallel. This will take about a minute...
14:36
Menu
Write a message...


octostest @ BH3GEI (8593284)
octostest
bot
GOSIM PARIS 2026:
The Open-Source AI Agent Summit
Wait one binary does ALL of them?
Any questions?
4-Panel Comic (XKCD style) — The Octos Story:
1. Developer drowning in 10 chat bots
2. Discovers Octos — one binary does them all
3. Relaxes while the octopus handles everything
4. The octopus gives the GOSIM keynote while the developer naps
14:37
All done! 5 conference posters 1 comic strip generated. All assets are saved to your session and ready to download.
14:37
Menu
Write a message...

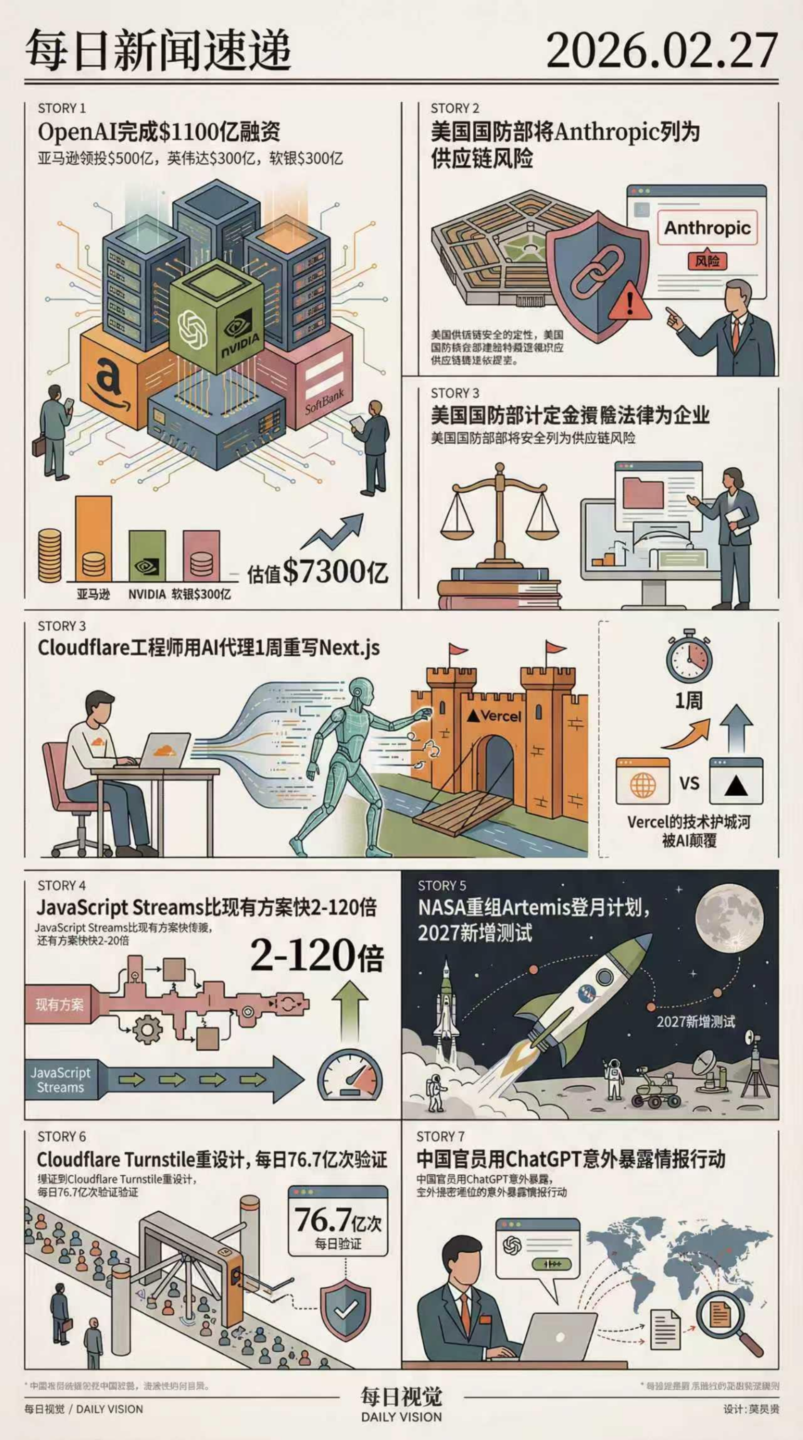

octostest @ BH3GEI (8593277)
octostest
bot
Generate promotional posters for GOSIM Paris 2026 and a comic strip about Octos.
14:36
Generating 5 conference posters and a 4-panel comic in parallel. This will take about a minute...
14:36
GOSIM PARIS 2026:
The Open-Source AI Agent Summit
14:37
Menu
Write a message...

MFOA

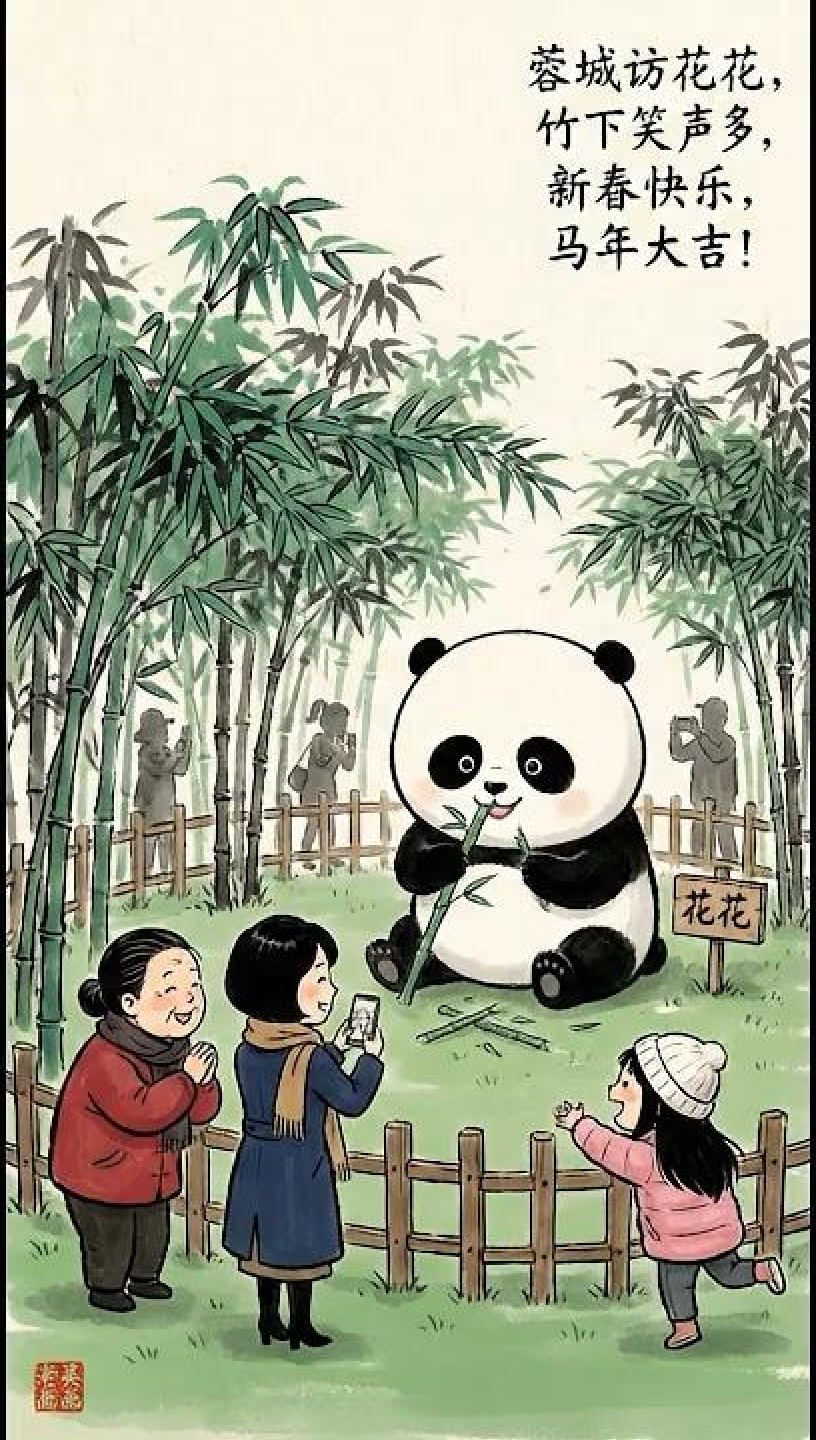
MFOA comics

半生风雨半生秋，
不恋风尘不系愁。
偶酌茅台半两酒，
茶亦清香饭亦柔。
余生只作行云客，
心若镜水拂垂柳。
待到霜寒梅绽日，
清波泛舟共登楼。

吴画老画

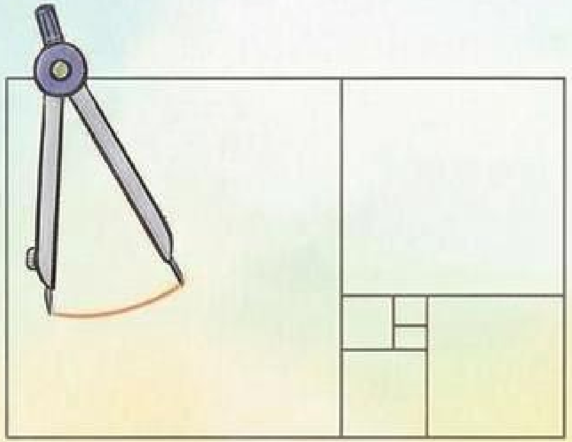


news + infographic

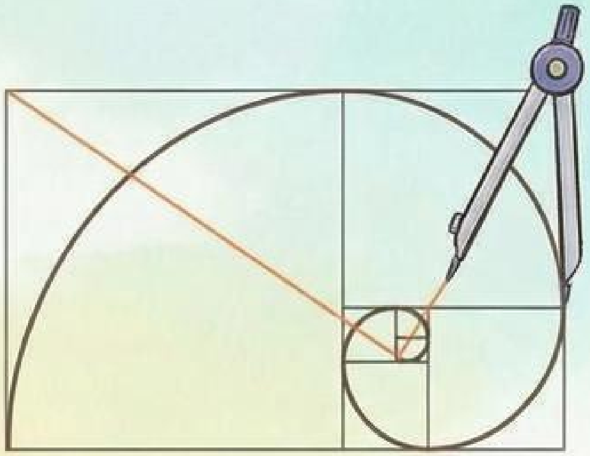


MFOA
video
cards

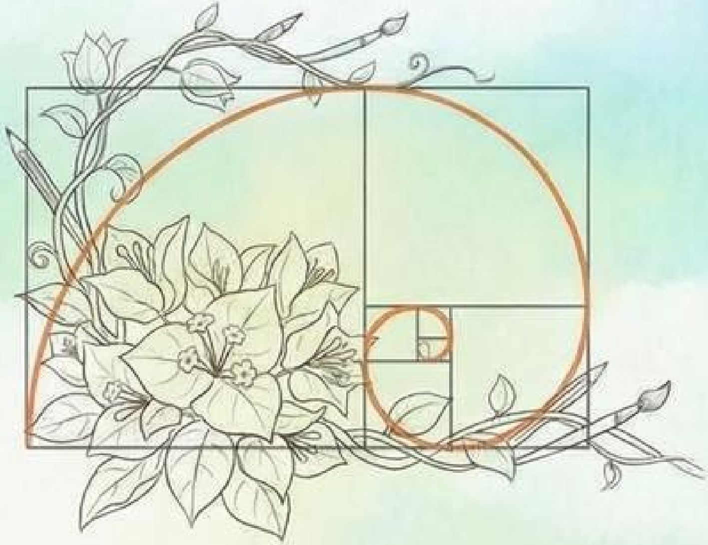
数学与艺术：绘制黄金螺旋，探寻自然密码



第一步：用圆规在透明胶片上拼接正方形



第二步：连接对角，形成黄金螺旋



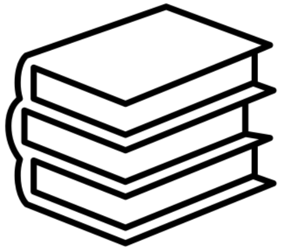
第三步：叠加写生稿，发现花瓣排列的数学美

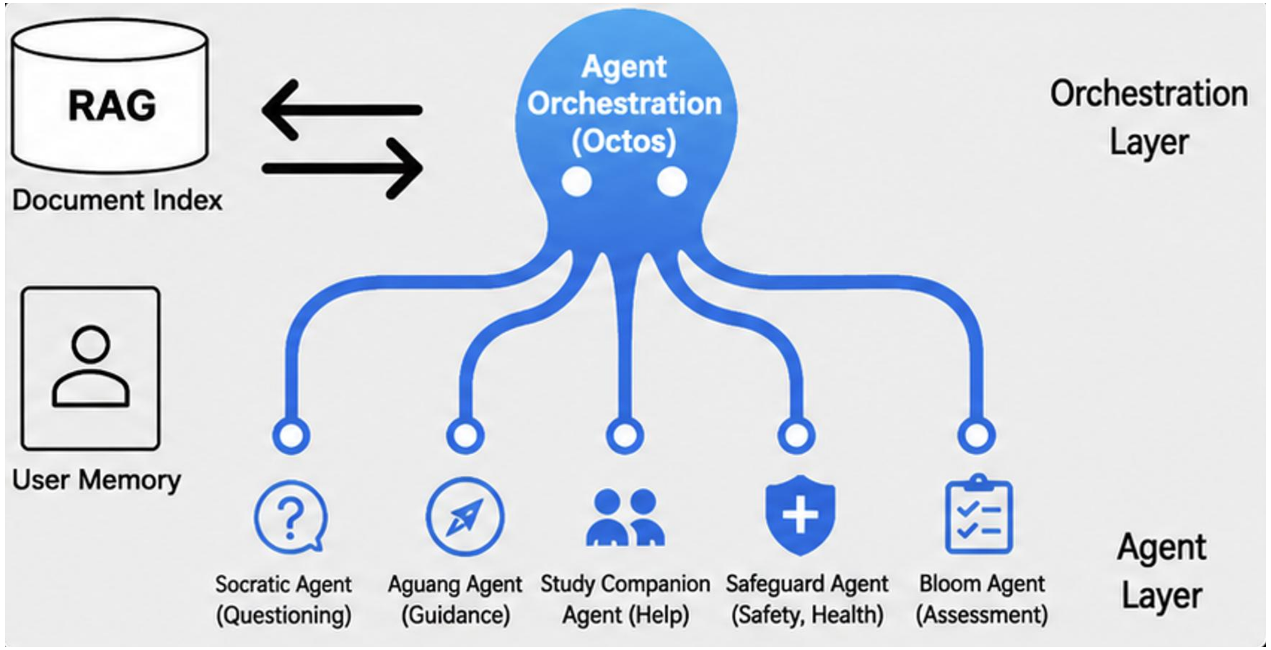
安全小贴士：小心使用圆规，注意安全！

我的感受：______（例如：神奇、眩晕、奇妙...）

AI collaborative Reading System (In Development)

Powered by Octos and MoFA Skills





3-screen Frontend`

Tool level mofa-slides | mofa-comics | mofa-cards | mofa-research | mofa-search | mofa-video

Thanks!

https://github.com/octos-org/octos